Sample Agency

Structural Design and Test Requirements for Launch Vehicles

Foreword

This sample standard is used only to test the chunking script.

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1 Scope

This standard establishes structural design requirements for launch vehicle hardware.

2 Applicable Documents

The documents listed in this clause are applicable to the extent specified herein.

3 Definitions

Ultimate load is the limit load multiplied by the ultimate factor of safety.

4 Requirements

4.1 Factors of Safety

Metallic structures shall have a minimum ultimate factor of safety of 1.4.

The following fastener table applies:

3 Bolts M6 torque 12 Nm

Composite structures shall use a factor of 1.5 unless approved by the project.

4.2 Pressure Vessels

Pressure vessels shall be proof tested to 1.5 times maximum expected operating pressure before flight.

4.2.1 Leak Testing

Each vessel shall be leak tested after proof testing.

Annex A Test Levels

A.1 Random Vibration

Qualification random vibration levels shall be 3 dB above acceptance levels.